

About

Senior Software Engineer with more than 20 years of experience building and evolving business-critical systems across telecommunications, payments, insurance, retail, energy, and application security.

I work across the full software delivery lifecycle — from solution architecture and backend services to mobile and web applications, automated testing, CI/CD, observability, and production support.

At Bosonit, working on a long-term project for TIM Brasil, I designed and continue to evolve an end-to-end preventive-maintenance platform. My responsibilities include the React Native application, Java backend, solution architecture, Jenkins pipelines, and ongoing product evolution.

Previously, at BugScout, I worked on static code analysis and application security, integrating Jenkins, Maven, SonarQube, SAP, and Docker. I developed Java and Spring Boot components, bytecode instrumentation, Java agents, and .NET test adapters, while also contributing to documentation, packaging, and front-end modernization.

My core expertise includes Java, Spring Boot, React Native, React, Node.js, Docker, and CI/CD. More recently, I have been expanding my work in applied AI, including RAG, AI agents, LangChain, LangGraph, OpenAI APIs, Google Cloud, and agent-native systems.

I am at my best when turning complex operational problems into reliable, maintainable products and owning the solution from architecture through production.

Highlights

- End-to-end ownership, from architecture to production
- Java, Spring Boot, React Native, React, Node.js, Docker, and CI/CD
- Business-critical systems and complex corporate integrations
- Application security and low-level code analysis
- Applied AI, AI agents, RAG, and agent-native systems

Languages

- Portuguese (Native)
- English (Professional)
- Spanish (Professional)

Education

Bachelor in Information Systems

Universidade Bandeirantes

2002 - 2006

MBA in Software Engineering with AI

FCTech - Full Cycle College

2025 - Present

Skills & Competencies

Applied AI & Agent-Native Systems

- AI agents and human-agent workflows
- Retrieval-Augmented Generation (RAG)
- LangChain / LangGraph
- OpenAI APIs and agent clients
- Model Context Protocol (MCP)

Architecture & Development Practices

- End-to-end software development lifecycle
- REST APIs (development, consumption, Swagger documentation)
- SOAP Web Services
- Secure development patterns
- Event-driven architecture
- Containerized environments (Docker, Docker Compose, Kubernetes)
- Microservices
- Low-level code analysis (bytecode)
- SonarQube plugin development
- Jenkins plugin development
- JVM agent development
- .NET Test Adapter development

Backend Languages & Frameworks

- Java
- Python
- Node.js

Testing & Quality Assurance

- JUnit
- Mockito
- TestContainers

●●●●● .NET / C#
●●●●● Spring / Spring Boot /
Spring Data
●●●●● Hibernate
●●●●● Quartz
●●●●● Kafka
●●●●● ActiveMQ
●●●●● Advanced Maven
configuration

●●●●● JMeter
●●●●● Postman

Mobile Development

●●●●● React Native
●●●●● Android (Java)
●●●●● Ionic
●●●●● Cordova

Frontend Development

●●●●● React.js
●●●●● Angular
●●●●● AngularJS
●●●●● HTML5 & CSS
●●●●● Bootstrap
●●●●● jQuery
●●●●● JSF / PrimeFaces

DevOps & Infrastructure

●●●●● Jenkins & Groovy
●●●●● Docker
●●●●● Tomcat
●●●●● JBoss
●●●●● WebLogic
●●●●● Apache
●●●●● Nginx
●●●●● Bash scripting
●●●●● Linux (Debian, CentOS)
●●●●● SonarQube
●●●●● Google Cloud
●●●●● Firebase
●●●●● Nexus

Data Management

●●●●● SQL Server
●●●●● Oracle
●●●●● MySQL
●●●●● Elasticsearch
●●●●● MongoDB
●●●●● Firestore
●●●●● Neo4J

Methodologies & Processes

●●●●● Scrum / Kanban
●●●●● Gitflow
●●●●● Code Review & Pull
Requests
●●●●● Requirements analysis

Selected Projects



Bondia Workspaces (BIAWS) — Creator & Lead Developer

Open-source, agent-native operational workspace for hybrid human and AI software teams. It brings together issues, improvements, operational procedures, application topology, runtime health signals, and versioned agent skills in a single platform.

Designed domain-scoped MCP tools and a CLI workflow for publishing and installing agent capabilities without unrestricted database access. The platform includes workspace-level authorization, audit trails, reproducible Docker installation, multi-instance support, and integrations with agent clients such as Codex and Claude.

Stack: Artificial Intelligence, Software Architecture, Node.js, React, MongoDB, Docker, MCP

GitHub repository (<https://github.com/fbondia/biaws>)



Manura — Creator & Software Architect

Modular platform for asset management, maintenance, and field operations. It supports asset lifecycle management, preventive, corrective and predictive activities, dynamic forms, configurable workflows, geographic planning, offline mobile execution, inventory, operational resources, audit trails, and AI-assisted configuration.

Responsible for product architecture across web, mobile, backend, shared domain components, automated testing, CI/CD, observability, and operational tooling.

Stack: Software Architecture, React Native, React, Node.js, Firebase, MongoDB, Playwright

Product website (<https://app.manura.co/>)



Quero QR — Creator & Full-Stack/Mobile Engineer

End-to-end platform that connects physical objects to persistent digital identities through stable QR Codes. Each item can have a structured digital profile with public or private information, photos, files, notes, checklists, history, collections, analytics, and visitor interactions.

Responsible for product design, software architecture, web and mobile development, backend services, billing, automated testing, deployment, and the complete physical-to-digital user experience.

Stack: React Native, React, Firebase, Google Cloud, Software Architecture, Playwright

Google Play (<https://play.google.com/store/apps/details?id=br.com.bondia.queroqr>)



I-Ching Mobile — Creator & Mobile Engineer

Independently designed and developed an Android application for I Ching consultations, with structured exploration of trigrams, hexagrams, changing lines, related hexagrams, and consultation history.

More recently, I added AI-assisted interpretation to help users explore consultation results from new perspectives while preserving the reflective, non-deterministic nature of the experience.

Stack: React Native, Firebase, Android, Mobile Architecture, Applied AI

Google Play (<https://play.google.com/store/apps/details?id=br.com.bondia.iching>)

Professional Experience

Senior Software Engineer / Solution Architect

TIM Brasil - 01/2014 - present

Designed and continue to evolve an end-to-end preventive-maintenance platform for TIM Brasil through Nalbatech/Bosonit.

Responsible for the React Native application, Java backend, solution architecture, corporate integrations, Jenkins pipelines, automated testing, production support, and continuous product evolution.

Technologies

- React.js, React Native
 - Java, Spring MVC, Hibernate
 - Ionic, Cordova
 - AngularJS, Bootstrap
 - Oracle
 - Linux Redhat and CentOS
 - Jenkins
 - JMeter
-

Senior Software Engineer

Bugscout Inc. - 08/2018 - 09/2024

Worked as a senior developer on a static code analysis system focused on information security (OWASP, CWE, PCI), integrating tools such as Jenkins, Maven, SonarQube, SAP, and Docker. Developed components in Java and Spring Boot, and worked with bytecode instrumentation, JavaAgents, and .NET TestAdapters. Also responsible for documentation, packaging, and redesigning the product's front end.

Technologies

- Java, Springboot, SpringData, SpringShell
- ActiveMQ, Kafka, Elasticsearch, Neo4J
- Swagger, REST API, GraphQL API, JWT
- React, Material UI
- Jenkins, Maven, Javac and Sonarqube plugins development
- Java and .NET low-level structures
- Java and .NET code instrumentation
- AST handling
- Bytecode/IL handling
- Development of TestAdapters (.NET)
- Development of JavaAgents (JVM)
- Development of Java x SAP integrations
- Kubernetes, Docker, Docker-Compose, TestContainers
- GCloud, AWS
- JUnit, Mockito
- Ansible, Packer
- Linux Debian, Bash scripting

Senior Developer

BNP Paribas Cardif - 02/2016 - 07/2018

Participated in the development of new features and maintenance of the company's core systems, such as those responsible for claims management automation and insurer-partner integrations.

Technologies

- Java
 - Hibernate
 - JBoss
 - Quartz
 - JPA
 - Spring-Batch
 - Sql Server
 - Angular 4
 - JSF Primefaces
-

Senior Software Developer

Schneider Electric - 01/2014 - 02/2016

Project developed through the consulting firm Nalbatech.

Worked on the development of a tool for calculating sales representative commissions.

Technologies

- .NET
 - C#
 - MVC
 - Oracle
 - jQuery
 - Bootstrap
-

Systems Specialist

Cielo - 06/2013 - 01/2014

Hired by Cielo as a Systems Specialist, working in the Low Platform Systems Support area. My role involved managing incidents, problems, requests, and improvements across a group of 35 company applications, including critical systems such as the one responsible for sending payment schedules to partner banks and batch transaction processing (EDI).

In addition to incident management, I oversaw software development vendors and served as a liaison between the technical team and internal departments such as Architecture, Testing, Infrastructure, and Change Management.

Technologies and Methodologies

- Itil v3
 - TN3270E
 - VB6
 - ASP.NET
 - Java + Spring MVC
 - C++
 - Linux
 - Sql Server
 - Oracle
 - SVN
-

Technical leader

Cielo - 01/2013 - 06/2013

After a period working for another company, I returned to Cielo to work in the support cell provided by H2M. A few months later, I was hired directly by Cielo, where I remained until January 2014.

Technologies

- VB6
 - ASP.NET
 - Java
 - Sql Server
 - Oracle
-

Consultant

Consultoria H2M - 08/2006 - 05/2013

Worked for H2M from 08/2006 to 06/2013. Participated in various projects for multiple clients, sometimes working on-site and other times in the software factory, where I also performed technical leadership roles.

Below are the projects I was involved in, listed from most to least recent:

Systems Analyst

Fundação Ezute - 10/2012 - 01/2013

Fundação Atech (now Ezute) is a Brazilian organization focused on developing and applying critical technologies. Worked through H2M on a system for managing water resource usage requests. A major takeaway from this project was learning to work with geospatial APIs (OpenLayers, OpenStreet, Leaflet).

Technologies

- C#
 - MVC
 - Sql Server
 - Entity Framework
 - OpenStreet
 - OpenLayers
 - LeafLet
 - Outsystems
-

Technical Lead

Cielo - 11/2009 - 09/2012

Projects developed through H2M.

Acted as a technical lead at Cielo, estimating efforts, drafting technical proposals and schedules, and coordinating project execution, engaging with both the client and the internal development team.

Technologies

- ASP.NET
 - C#
 - Java
 - VB6
 - Oracle
 - Sql Server
 - Ms-Project
-

Software Developer Analyst

Merrill Lynch - 11/2007 - 11/2009

Designed and developed a system used by both the Back and Front Office to manage the company's stock portfolio and lending operations with other brokerages.

Worked through H2M.

Technologies

- ASP.NET
 - MS-AJAX
 - XML
 - C#
 - Sql Server
 - Windows Services
-

Team Coordinator

C&A - 11/2008 - 01/2009

Project developed through H2M.

Coordinated the development of a system for managing the delivery and collection of change (cash) at C&A stores in Brazil, Argentina, and Mexico.

Technologies

- ASP.NET
 - C#
 - DB2
-

Software Developer Analyst

Banco Espirito Santo - 10/2008 - 11/2008

One-off project through H2M.

Developed a system to import data from various sources (Excel, Text, CSV, OleDB) and perform comparisons, thereby optimizing routine reconciliation processes.

Technologies

- AJAX
 - YUI
 - DHTML
 - ASP.NET
 - XML
 - C#
 - Sql Server
 - Windows Services
 - Windows Forms
-

Systems Analyst

Editora SBS - 09/2007 - 11/2007

SBS is a language-focused book publisher. Provided consulting services through H2M, mapping their logistics process and contributing to the creation of an automated model.

Software Developer Analyst

Drogacenter - 08/2006 - 11/2007

Provided services through H2M, developing an electronic order manager that monitored and integrated mobile-submitted orders into the ERP system.

Technologies

- C#
 - Windows Services
 - Windows Forms
 - XML
-

Software Developer Analyst

NationSoft - 07/2006 - 10/2007

Worked for approximately a year and a half developing various internal products, such as performance management systems, timesheet systems, data integration tools, résumé databases, among others.

Technologies

- AJAX
 - YUI
 - DHTML
 - ASP.NET
 - Web Services
 - XML
 - C#
 - Sql Server
-

Developer

Hewlett-Packard - 02/2002 - 07/2006

Developed applications to support HR processes at HP. Contributed to the department's intranet, creating self-service tools for employees.

Technologies

- ASP.NET
 - C#
 - XML
 - ASP
 - DHTML (JavaScript, Html, CSS)
 - VB 6.0
 - VBA
 - Oracle
 - PL/SQL
-

Administrative Assistant

Hewlett-Packard - 10/1996 - 02/2002

Worked for almost six years in the Human Resources administrative area – Personnel Administration – responsible for onboarding processes for employees and interns.

Despite being in an administrative area, I developed several tools to optimize tasks. Eventually, I was invited to join the IT department responsible for HR technology.

Technologies

- ASP.NET
 - MS Access
 - MS Office
 - VBA
 - VB6
 - Oracle
 - PeopleSoft
-